



Perception-to-Image: Reconstructing Natural Images from the Brain Activity of Visual Perception

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Abstract—The reappearance of human visual perception is a challenging topic in the field of brain decoding. Due to the complexity of visual stimuli and the constraints of fMRI data collection, the present decoding methods can only reconstruct the basic outline or provide similar figures/features of the perceived natural stimuli. To achieve a high-quality and high-resolution reconstruction of natural images from brain activity, this paper presents an end-to-end perception reconstruction model called the similarity-conditions generative adversarial network (SC-GAN), where visually perceptible images are reconstructed based on human visual cortex responses. The SC-GAN extracts the high-level semantic features of natural images and corresponding visual cortical responses and then introduces the semantic features as conditions of generative adversarial networks (GANs) to realize the perceptual reconstruction of visual images. The experimental results show that the semantic features extracted from SC-GAN play a key role in the reconstruction of natural images. The similarity between the presented and reconstructed images obtained by the SC-GAN is significantly higher than that obtained by a condition generative adversarial network (C-GAN). The model we proposed offers a potential perspective for decoding the brain activity of complex natural stimuli.

Keywords—Visual decoding, Reconstruction, SC-GAN, Deep learning.

INTRODUCTION

Visualizing the perception of the human brain is a challenging goal in neuroscience, and brain decoding methods using machine learning based on fMRI activities have made the visualization of perceptual content possible.¹⁷ The reconstruction of natural image perception, which aims to map a visual cortex response space to an image feature and semantical space, is an important research topic in the field of brain decoding. The reconstructed images should ideally preserve the low-level image features as well as the high-level semantic information of the original targets.

In recent years, generative adversarial networks (GANs⁵), as deep learning models, have been widely used in various research fields such as image conversion,²⁰ image synthesis,¹⁵ image restoration,² image super-resolution,¹⁰ text-to-image synthesis,²¹ *etc.* Recently, several studies have used generative adversarial networks (GANs) for the reconstruction of facial images⁶ and natural images.^{16–18} For example, Güçlütürk *et al.* presented a deep adversarial neural decoding approach to reconstruct perceived stimuli from brain responses by combining probabilistic inference with deep learning.⁶ The approach first performed a linear transformation from latent features to brain responses with maximum *a posteriori* estimation and then implemented a nonlinear transformation from perceived stimuli to latent features with the adversarial training of convolutional neural networks. Although the deep adversarial neural decoding approach used an adversarial training approach, it

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cannot be considered a real GAN-based brain decoding method. Seeliger *et al.* developed a straightforward method for reconstructing visual stimuli directly from brain activity by pre-training a deep convolutional generative adversarial network.¹⁶ They achieved the reconstruction of grayscale images with fMRI signals. Ghislain *et al.* applied a GAN to learn a generative model of images that was conditioned on measurements of brain activity and then achieved the reconstruction of the basic outline of the stimuli.¹⁸ Inspired by the research that visual cortical activity can be decoded (translated) into the hierarchical features of a deep neural network (DNN) for the same input image, Shen *et al.* proposed a deep image reconstruction model to visualize the perceived content from human brain activity.¹⁷ The model also made use of an *a priori* deep generator network (DGN) to constrain the reconstructed images and make the reconstructed images similar to the original ones. The study found that there was no significant difference in reconstruction accuracy with or without a DGN.

Although some advances in GAN- or C-GAN-based brain decoding have been made by the above studies, the present methods can only reconstruct the basic outline or provide similar figures/features of the perceived natural stimuli. High-quality and high-resolution reconstruction from brain activity still has a long way to go. The main difficulty in accurately reconstructing natural images lies in the fact that neural representation is noisy, highly dimensional, and it contains incomplete information about image details.¹³ Massive amounts of neuroimaging data are required to characterize the invariances of visual features and establish the relationship between neural representations and image features.⁷ However, due to the constraints of fMRI data collection, the sample size is always limited.¹ Therefore, enhancing the mapping between neural response and image features is a key issue for improving reconstruction.

In this paper, we proposed a strategy of disguising the fMRI sample size to solve the insufficient training sample problem during natural image reconstruction. An end-to-end perception reconstruction model, a similarity-conditions generative adversarial network (SC-GAN), is proposed to directly extract visual perception information from human brain activity and reconstruct natural images. The SC-GAN includes an image feature extractor, a brain feature extractor, a generator and a discriminator. To solve the problem of insufficient sample size from the visual perception space to the colorful image space, we make use of additional information, namely, whether the stimulus image is paired with corresponding brain activity or not, to constrain the brain feature extractor and extract semantic representation features of visually per-

ceived images. The semantic representation feature is input synchronously to the generator and discriminator as a condition to reconstruct the perceived natural image. Based on C-GAN, we added an image feature extractor and brain feature extractor to construct the SC-GAN and defined a similarity loss function to extract latent brain features. The latent brain features obtained by the brain feature extractor are input as conditions to the model to produce the reconstructed images. We validate our proposed SC-GAN with a visual perceptual experiment. During the experiment, five types of images (horses, buildings, flowers, fruits and landscapes) were shown to subjects with random intervals, and fMRI responses were collected while the subjects viewed the images. The experimental data were divided into training and test sets for training and testing the framework. Compared with previous reconstruction model based on C-GAN, the similarity index between the presented and reconstructed images obtained by the SC-GAN was significantly improved.

MATERIALS AND METHODS

Subjects and Experimental Design

Five subjects (3 males and 2 females, age 23–27 year, average 25) participated in this experiment. All subjects had no eye diseases and had normal or corrected-to-normal vision. Written informed consent was obtained from each subject before the experiments. The experimental protocol was approved by the Institutional Review Board of the Institute of Biophysics, Chinese Academy of Sciences. During the experiment, subjects were instructed to focus on the center of the screen without moving their bodies.

The experiment included two sessions: a bar retinotopic mapping session³ and a natural image presentation session. The bar retinotopic mapping session was performed to delineate visual areas. In this session, drifting bar apertures were presented to the subjects in the form of a checkboard pattern with 100% contrast (spatial length: 20°; spatial width: 20°; temporal frequency: 10 Hz). There were eight bar configurations within a given scan (four bar orientations and two different motion directions), and each bar configuration had twenty-two equidistant spatial positions. Four repeated runs were performed in this session, and each run contained of 176 trials. In each trial, the bar in a specific position flashed for 2 s and was then followed by the next trial. At the beginning and the end of each run, a gray background was displayed for 12 s as the rest periods.

In the natural image presentation session, subjects were shown 2750 natural images that belonged to five

categories (horses, buildings, flowers, fruits and landscapes). The natural images were obtained from the image set, which was used with a resolution of 256×256 . There were 550 natural images in each image category. Fifty-five runs were performed in this session. In each run, 50 natural images (10 images from each category) were presented in pseudorandom order with a 2 s presentation time and a 4–8 s random interval. The images were randomly selected without repetition. The gray background with a central white fixation cross was presented for 12 s at the beginning and 8 s at the end of each run. Fifty and five runs were randomly selected as the training and test sets, respectively.

Data Acquisition

MRI data were acquired at the Institute of Biophysics, Chinese Academy of Sciences using a 3-T Prisma^{fit} scanner (Siemens, Erlangen, Germany) and a 20-element head-neck coil. To acquire functional data that cover the entire occipital lobe, an interleaved multiband T2*-weighted gradient-echo echo-planar imaging (EPI) pulse sequence was used (TR, 1000 ms; TE, 31.2 ms; multiband, 4; FOV, 194×194 mm²; flip angle, 50°; spatial resolution, $1.8 \text{ mm} \times 1.8 \text{ mm} \times 1.8$ mm; number of slices, 48; slice gap, 0 mm). For anatomical data, a T1-weighted magnetization-prepared rapid-acquisition gradient-echo (MP-RAGE) was used (TR, 2300 ms; TE, 3.49 ms; TI, 1050 ms; FOV, 256×256 mm; flip angle, 8°; spatial resolution, $1.0 \text{ mm} \times 1.0 \text{ mm} \times 1.0$ mm).

Data preprocessing was performed with SPM8 (<http://www.fil.ion.ucl.ac.uk/spm>). The first 12 volumes of each run were discarded to allow for a magnetic saturation effect.⁴ Subsequent images were corrected by slice timing and head motion. Then, fMRI images were coregistered to the sliced high-resolution anatomical data. Finally, fMRI data from each run underwent z-score transformation by subtracting the mean and being divided by the standard deviation of the time series signal at each voxel.⁹

After the bar retinotopic mapping session, the method of estimating the neuronal population receptive field (pRF) in the human visual cortex was applied to label the visual region of interest (ROI) based on a previous study.³ Voxels in V1, V2, V3 and the remaining area of VC (defined as the high visual cortex (hVC)) were combined together to train and test the decoder. In the training data set, F-score feature selection⁸ was used to screen voxels with high response values for each visual cortex (V1, V2, V3 and hVC) and retain the first 2000 voxels for subsequent analysis.

Perception Reconstruction Model Based on SC-GAN Framework

The SC-GAN model consists of four submodules: image feature extractor, brain feature extractor, generator, and discriminator, as shown in Fig. 1. The image feature extractor and brain feature extractor encode the image and visual response patterns into latent image features and latent brain features, respectively. The cosine similarity algorithm is used to calculate the similarity between latent image features and latent brain features. Based on the similarity of the paired or unpaired calculations, the similarity loss function is proposed as follows:

$$L_1 = C - F_{CS}(f_{IFE}(x), f_{BFE}(y)) + F_{CS}(f_{IFE}(x), f_{BFE}(\tilde{y})) \quad (1)$$

$$L_2 = C - F_{CS}(f_{IFE}(x), f_{BFE}(y)) + F_{CS}(f_{IFE}(\tilde{x}), f_{BFE}(y)) \quad (2)$$

$$L_{\text{similarity}} = \begin{aligned} & E_{x, \tilde{x} \sim P_{\text{img}}; y, \tilde{y} \sim P_{\text{fmri}}} [\max(0, L_1)] \\ & + E_{x, \tilde{x} \sim P_{\text{img}}; y, \tilde{y} \sim P_{\text{fmri}}} [\max(0, L_2)] \end{aligned} \quad (3)$$

where x, y denote an image and the corresponding visual response patterns, respectively. The image (x) and the corresponding visual response patterns (y) form a paired sample. $\tilde{x}$ denotes an image randomly selected from the image set in which the x image is removed. $\tilde{y}$ denotes the visual response patterns randomly selected from the visual response patterns set in which the y is removed. f_{IFE} , f_{BFE} denote an image feature extractor and the brain feature extractor, respectively. F_{CS} denotes the cosine similarity function. C represents a non-zero decimal, which makes the parameters in the image feature extractor and the brain feature extractor always updatable during training. Here, we refer to the common design of leaky-ReLU, the value of C is set as 0.2.

In addition, a C-GAN consisting of a generator G and a discriminator D is adopted. The latent brain features are used as conditions for the C-GAN and are simultaneously input into the generator and discriminator networks. The objective function of the C-GAN is as follows:

$$\begin{aligned} \min_G \max_D V(D, G) = & E_{z \sim N(0,1); y \sim P_{\text{fmri}}} [\log(1 - D(G(z, f_{BFE}(y)), f_{BFE}(y)))] \\ & + E_{x \sim P_{\text{img}}; y \sim P_{\text{fmri}}} [\log(1 - D(x, f_{BFE}(y)))] \\ & + E_{x \sim P_{\text{img}}; \tilde{y} \sim P_{\text{fmri}}} [\log(D(x, f_{BFE}(\tilde{y})))] \end{aligned}$$

where $z \sim N(0, 1)$ denotes random noises.

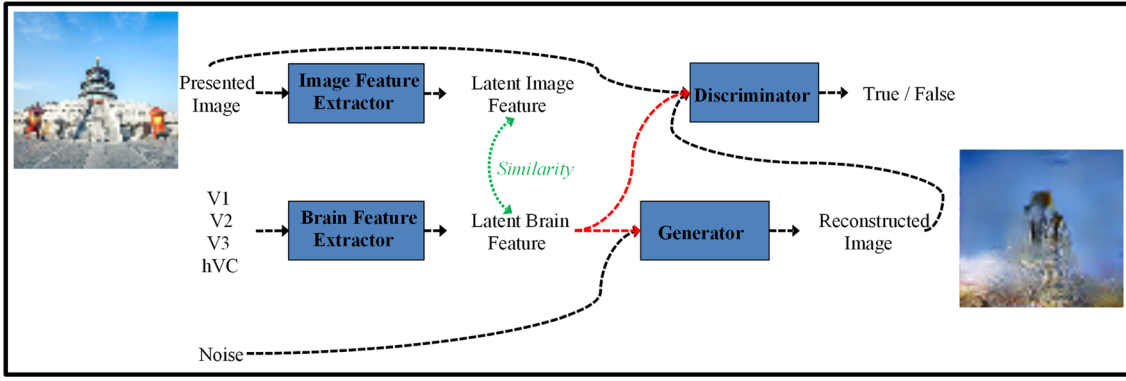


FIGURE 1. Schematic diagram of the similarity-conditions generative adversarial network for natural image reconstruction.

Note that the similarity loss function is only used for updating the parameters in the image feature extractor and the brain feature extractor. The objective function of C-GAN is only used for updating the parameters in generator and discriminator. The final objective function of SC-GAN consists of two parts: the similarity loss function and the loss function of C-GAN. The two loss functions are synchronized during training. The significance of the similarity loss function is to optimize the parameters in the image feature extractor and the brain feature extractor by matching or mismatching the image and the visual response patterns. The aim is to extract the effective latent brain features from visual response patterns for reconstruction. The latent features extracted in this way contain common valid information in both the image and the visual cortex patterns. The loss function of C-GAN is designed to reconstruct similar images according to similar conditions (latent brain features).

We choose x , $\tilde{x}$, y , and $\tilde{y}$ from the training sample set. x and $\tilde{x}$ are natural images. y and $\tilde{y}$ are visual response patterns. Among them, x and y are paired. $\tilde{x}$ and $\tilde{y}$ are randomly selected from the natural image set and the visual cortex patterns set, respectively. Therefore, $\tilde{x}$ - y and x - $\tilde{y}$ are unpaired. During the training stage, we update the parameters of SC-GAN as follows:

- (1) With image feature extractor, brain feature extractor, and cosine similarity algorithm, we can construct a similarity loss function. The similarity loss function and Adam optimizer are used to update the parameters in image feature extractor and brain feature extractor.
- (2) With generator and discriminator, we can construct a loss function of the C-GAN. The loss function of the C-GAN and Adam optimizer are used to update the parameters in generator and discriminator.

The default parameter settings of the Adam optimizer (beta1 = 0.5, beta2 = 0.999, epsilon = 1e-8) and the learning rate (0.0001) are used. Here, the SC-GAN is implemented in TensorFlow. During the test stage, we use the visual response patterns (from test set), the brain feature extractor, and the generator to obtain the reconstructed images.

Architecture of the Image Feature Extractor

In the image feature extractor, RGB images with dimensions [64, 64, 3] are first mapped using convolution (kernel: $3 \times 3 \times 64$; step size: 2) and leaky-ReLU (0.2) operations to features with dimensions [32, 32, 64]. Immediately thereafter, these features are sequentially operated through convolution (step size: 2), batch normalization and leaky-ReLU (0.2) steps, which are repeated three times, and mapped to features with dimensions [4, 4, 512]. The kernels are $3 \times 3 \times 128$, $3 \times 3 \times 256$, and $3 \times 3 \times 512$. Finally, these features are mapped to latent image features with dimensions [1024] by flat and fully connected operations. The network structure of the image feature extractor is shown in Fig. 2.

Architecture of Brain Feature Extractor

In the brain feature extractor, the visual response patterns (V1, V2, V3, hVC) with dimensions [4, 2000] are first mapped using fully connected, batch normalization and leaky-ReLU (0.2) operations to the features with dimensions [4, 2048]. Next, the feature is mapped to the [4, 1024] dimension using the LSTM operation. Only the last dimension is retained as a new feature with dimensions [1024]. Finally, the feature process is repeated four times in sequence through fully connected, batch normalization and leaky-ReLU (0.2) operations. Among them, the batch normalization and leaky-ReLU (0.2) operations are removed from the last

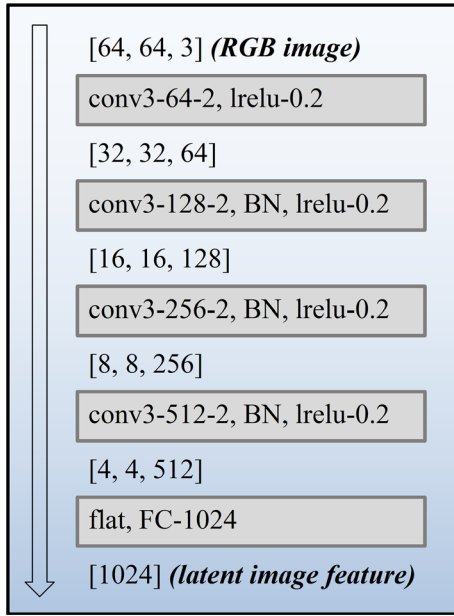


FIGURE 2. Architecture of image feature extractor.

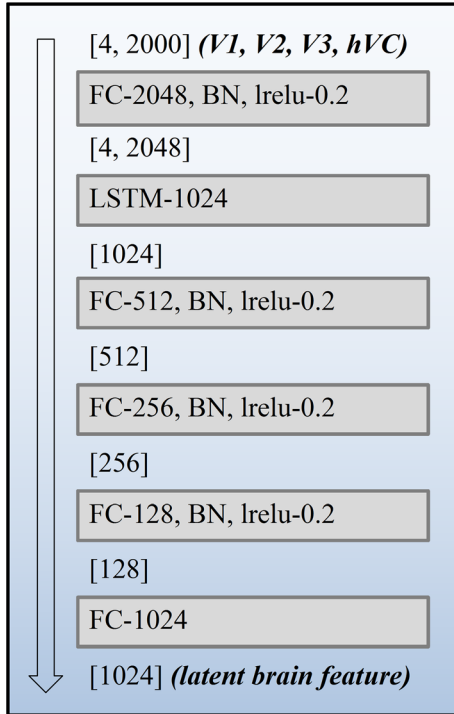


FIGURE 3. Architecture design of brain feature extractor.

cycle. The features are mapped to the latent brain feature space with dimensions [1024]. The network structure of the brain feature extractor is shown in Fig. 3.

Generator Network Architecture

In the generator, the latent brain features are first mapped using a fully connected operation to intermediate features with dimensions [1024], followed by leaky-ReLU (0.2), and are then concatenated to the noise vector z with dimensions [512]. Next, the concatenated features are mapped to features with dimensions [4, 4, 1024] by fully connected, batch normalization, and leaky-ReLU (0.2) operations. These features then pass through a residual module with the feature dimensions unchanged. The residual module in turn consists of repeating the convolution (step size: 1), batch normalization, and ReLU operations twice, followed by the add operation. The kernels are $1 \times 1 \times 256$ and $3 \times 3 \times 1024$. Then, the feature dimensions are transformed into [8, 8, 512] by the upsampling, convolution (kernel: $3 \times 3 \times 512$; step size: 1) and batch normalization operations. Next, the features pass through a residual module with the feature dimensions unchanged. The residual module in turn consists of repeating the convolution (step size: 1), batch normalization and ReLU operations three times, followed by the add operation. The kernels are $1 \times 1 \times 128$, $3 \times 3 \times 128$ and $3 \times 3 \times 512$. Finally, the features are repeated three times in sequence through unsampling, convolution (step size: 1), batch normalization and ReLU operations. Among them, the batch normalization and ReLU operations are removed at the last time. The kernels are $3 \times 3 \times 256$, $3 \times 3 \times 128$ and $3 \times 3 \times 3$. The features are mapped to an image space with dimensions of [64, 64, 3]. The network structure of the generator is shown in Fig. 4.

Discriminator Network Architecture

In the discriminator, the RGB images with dimensions [64, 64, 3] are first repeated four times in sequence through convolution (step size: 2), batch normalization and leaky-ReLU (0.2) operations. The kernels are $3 \times 3 \times 64$, $3 \times 3 \times 128$, $3 \times 3 \times 256$ and $3 \times 3 \times 512$. The feature dimension is transformed into [4, 4, 512]. These features then pass through a residual module with the feature dimensions unchanged. The residual module in turn consists of repeating the convolution (step size: 1), batch normalization, and leaky-ReLU (0.2) operations three times, followed by the add operation. The kernels are $1 \times 1 \times 128$, $3 \times 3 \times 128$ and $3 \times 3 \times 512$. At the same time, the latent brain features are first mapped using fully connected, batch normalization and leaky-ReLU (0.2) operations to a feature with dimensions [512] and then tiled to the feature with dimensions [4, 4, 512]. Features from RGB images and latent brain features are stitched into the features with dimensions [4, 4, 1024]. Finally, the fea-

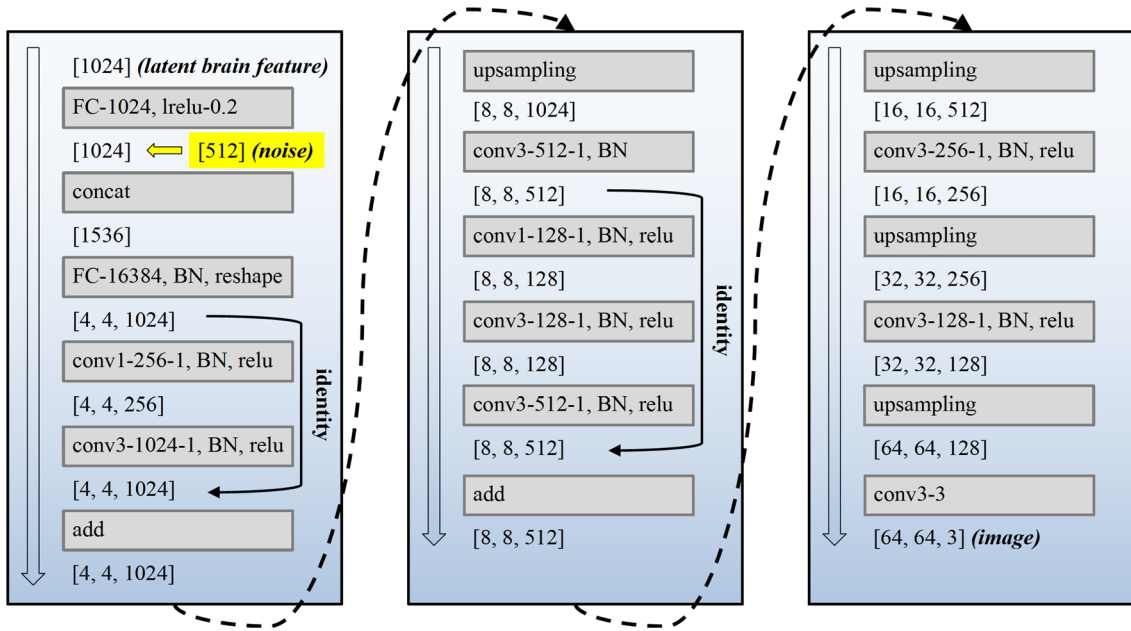


FIGURE 4. Generator network architecture.

tures are repeated two times in sequence through fully connected, batch normalization and leaky-ReLU (0.2) operations. Among them, the batch normalization and leaky-ReLU (0.2) operations are removed the last time. The features are mapped to a spot space with dimensions [1, 1, 1]. The network structure of the discriminator is shown in Fig. 5.

RESULTS

Reconstructed Images

In the experiment, when a subject viewed each natural image (a total of 2750 natural images), the response patterns of V1, V2, V3, and hVC were recorded by fMRI. The images and the visual response patterns (V1, V2, V3, and hVC) were matched into 2750 sample pairs. Each pair contains a natural image and the corresponding matrix of response patterns from four visual areas. These samples are randomly divided into training (2700 samples) and test (50 samples) sets. The training set is fed to the proposed SC-GAN for training to obtain model parameters. The test set is used to test the trained SC-GAN and obtain the reconstructed images from brain activities. The 50 test samples contained horses, buildings, flowers, fruits, and landscapes, with 10 samples per category. The training and the reconstruction are based on per-subject due to the individual difference of brain structure and perceptual activities. Totally five subjects are included in the experiment. Test stimuli and the corresponding reconstructed images from the visual

cortex responses of subject 1 are shown in Fig. 6. The results of the other four subjects are shown in Supplementary Figures 1–4. For each category, the upper row represents the stimulus images that the subject is viewing in the magnetic resonance machine, and the lower row represents the corresponding reconstructed images acquired from the visual response patterns by the SC-GAN. From Fig. 6, the stimuli images can be substantially reproduced from the visual response patterns. The results show that most of the reconstructed images have a certain degree of similarity with the presented images. It can also be found that the reconstruction effect of fruits with simple background and color is relatively better than other categories. Due to the complex targets and complicated backgrounds, the reconstruction of the other four categories of images is slightly worse than that of a fruit category. To illustrate the effectiveness of the SC-GAN, random signals (0–1 uniform distribution) are used to replace the visual response patterns and obtain the corresponding reconstructed images, shown in Supplementary Fig. 5. The results show that the reconstructed images obtained using the random signals show no visual similarity to the presented images. In addition, a C-GAN model is also used to compare with the SC-GAN, and the reconstructed image from subject 1 is shown in Supplementary Figure 6. Compared with the SC-GAN, the C-GAN has no image feature extractor, but the remaining structures and parameters are the same as the SC-GAN. The results of reconstruction using the C-GAN show that the presented stimulus images cannot be effectively reconstructed. In general,

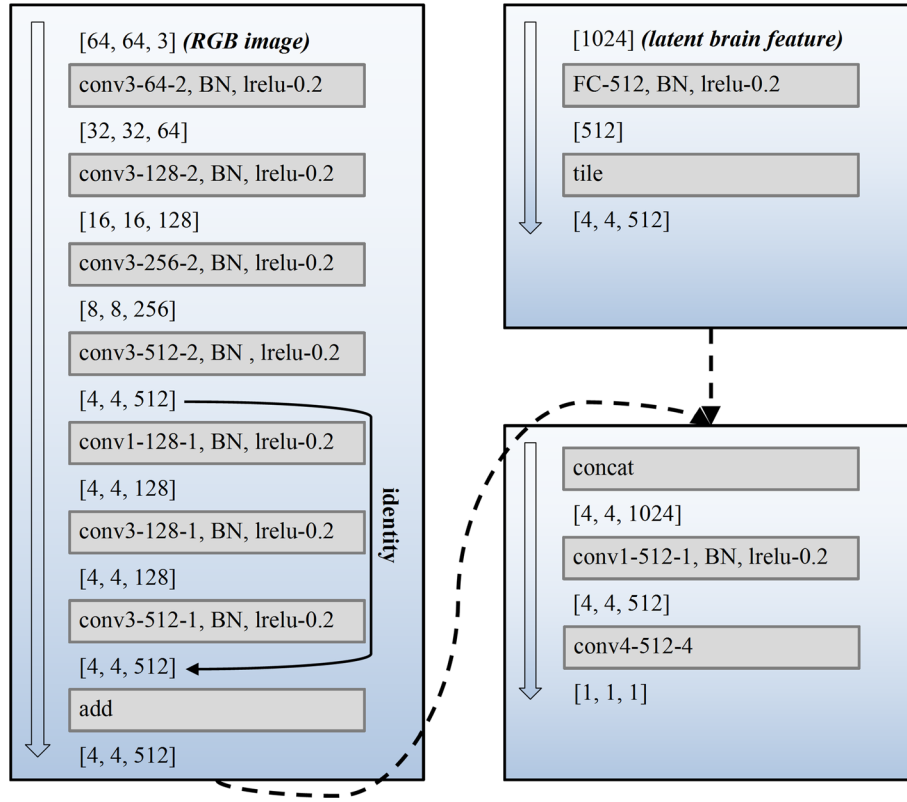


FIGURE 5. Discriminator network architecture.

our proposed SC-GAN can achieve a better reconstruction of the presented stimulus image from the visual response pattern.

Quality Assessment of Reconstructed Images

Quantitative measurements are performed to illustrate the performance of our proposed SC-GAN for reconstructing presented images from visually perceptual signals. Here, four methods are used to calculate the similarity between the reconstructed and the presented images. These methods are mutual information similarity (MI-SIM¹⁴), histogram similarity (H-SIM¹¹), structural similarity (S-SIM¹⁹) and cosine similarity (C-SIM¹²). The mean reconstruction similarity curves with the increasing of the iteration from the visual response patterns of the five subjects by four methods are shown in Fig. 7. The red curves represent the similarity obtained by feeding the visual response patterns into SC-GAN. The blue curves represent the similarity obtained by feeding the visual response patterns into C-GAN. The green curves represent the similarity obtained by feeding the random signals into SC-GAN. The results show that the reconstruction similarity from the brain activities of the five subjects is obviously higher than that from random signals. In addition, the results also show that the reconstruction

similarity of the proposed SC-GAN is significantly higher than that of the C-GAN. Except for the curves obtained from the random signal and the C-GAN, the other curves rise with the iterative process. It can be concluded that our proposed SC-GAN can learn effective information from visual response patterns and obtain good reconstructed images. In addition, a paired sample t test (See Supplementary Annotation) was used to test the significant difference between the similarity curves obtained from the visual response patterns and the random signal as well as the C-GAN model. The results show that the similarity curve obtained from the visual response patterns is significantly higher ($p < 0.01$) than that obtained from the random signal, and the similarity curve obtained by the SC-GAN is also significantly higher ($p < 0.01$) than that obtained by the C-GAN.

DISCUSSION

In this paper, we presented a new approach, the similarity-conditions generative adversarial network (SC-GAN), to reconstruct natural images based on visually perceptual signals from human brain activity. From both intuitive and quantitative points of view, we proved that the reconstructed images obtained by

Reconstructed results (subject 1)

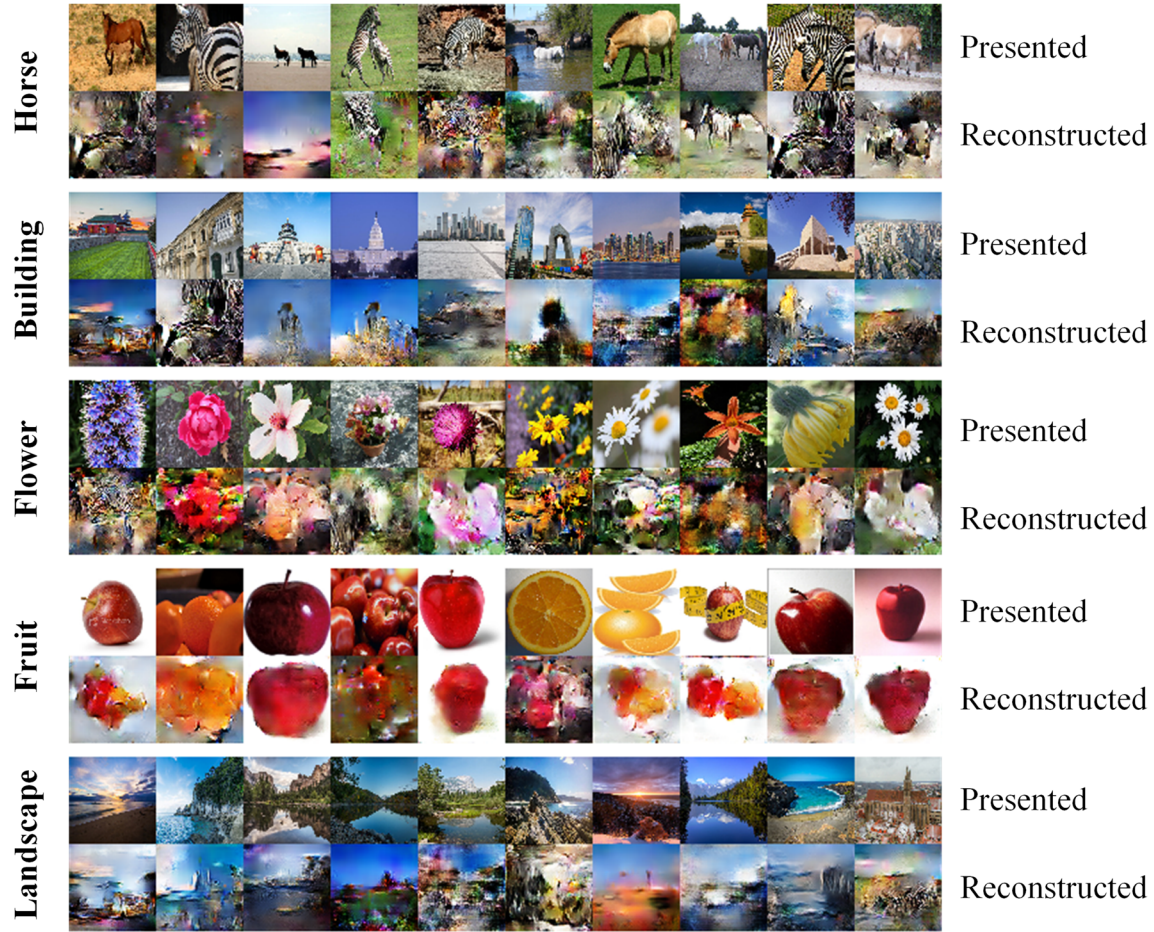


FIGURE 6. Reconstructed samples of the five categories of test stimuli (horse, building, flower, fruit and landscape) from the visual cortex responses of subject 1. The top row of each category represents the test images that the subject viewed during fMRI data recording. The bottom row of each category represents the corresponding reconstructed images from the brain responses.

this method show comparable similarity with the presented image. There are two contributions in this research. On one hand, the proposed SC-GAN has a reasonable architecture, which consists of an image feature extractor, a brain feature extractor, a generator and a discriminator. The loss function defined by the similarity obtained by matching or mismatching between the images and brain activities is used to synchronously train the image feature extractor and the brain feature extractor to extract effective information of the visually perceived content. This training method is different from previous research.^{7,17} For example, the Shen team (1) first extracted the latent features of natural images through pre-trained convolutional neural networks or other algorithms, and (2) then established a predictive model from visual response patterns to latent features through sparse regression models. They had to pre-train large convolutional neural network

models and build predictive models for each feature, which is not an optimal method. Here, we designed an image feature extractor and a brain feature extractor to extract the common information of visual response patterns and visual images more easily and more conveniently.

On the other hand, our proposed model makes full use of additional information, namely, the matching or mismatching pairs between the stimulus image and brain activity, to solve the insufficient training sample problem during natural image reconstruction. Take the green lines in Fig. 7 for example, because there is no correlation between the random signals and the natural images (mismatching condition between signals and images), the model cannot extract effective high-level semantic information from the signals, resulting in the total failure of effective reconstruction of the images (See Supplementary Figure 5). On the contrary, if the brain activities have correlation with the presented

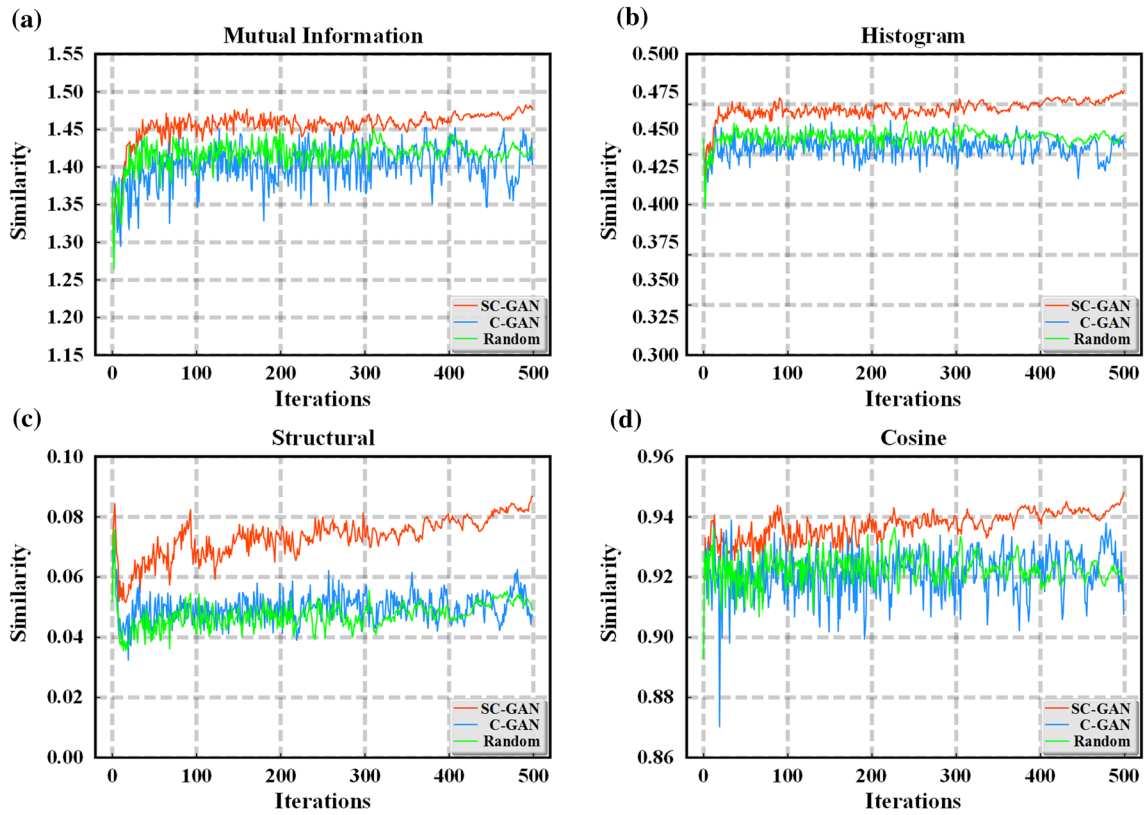


FIGURE 7. Mean reconstruction similarity curves between the reconstructed and the presented images with the increasing of the iteration from the visual response patterns of the five subjects by four methods. The red curves represent the mean similarity obtained from the visual response patterns by the SC-GAN. The blue curves represent the mean similarity obtained by the C-GAN. The green curves represent the mean similarity obtained from random signals by the SC-GAN. (a) The mean similarity curves by the mutual information similarity. (b) The mean similarity curves by the histogram similarity. (c) The mean similarity curves by the structural similarity. (d) The mean similarity curves by the cosine similarity.

images (matching condition between signals and images), the image feature extractor and the brain feature extractor can extract effective information of the visually perceived images, so higher reconstruction accuracy can achieve (red lines in Fig. 7). By introducing mismatching samples, the training sample size increases and the performance of the reconstruction is enhanced. Obviously, the visual response patterns evoked by natural images contain useful semantic information related to natural images as well as other irrelevant information. In general, the signal-to-noise ratio of the brain response signals measured by fMRI is relatively low. From this, we can infer that there is far more irrelevant information in the visual response patterns than semantic information. If we do not use image feature extractor, brain feature extractor, and the similarity loss function to extract valid semantic information, the latent brain feature contains little semantic information. It is conceivable that the quality of the reconstructed image obtained by feeding the latent brain feature into C-GAN is unsatisfactory (See Supplementary Figure 6).

In conclusion, the proposed SC-GAN takes advantage of the image feature extractor, the brain activity feature extractor and the similarity loss function to extract effective features, besides, it is trained by matching and mismatching pairs of images and brain activities, so the quality of the reconstructed image obtained is higher than traditional C-GAN. Paired sample T-test shows that the similarity between the reconstructed and presented images obtained by the SC-GAN is significantly higher than that of the C-GAN. This fully demonstrates that our design of image feature extractor, brain feature extractor and similarity loss function can effectively extract the latent brain features. This unique network design offers a potential perspective for better reconstructing natural images from brain activity responses.

ELECTRONIC SUPPLEMENTARY MATERIAL

The online version of this article (doi:<https://doi.org/10.1007/s10439-020-02502-3>) contains supplementary material, which is available to authorized users.

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REFERENCES

- ¹Button, K. S., J. P. Ioannidis, C. Mokrysz, B. A. Nosek, J. Flint, E. S. Robinson, and M. R. Munafò. Power failure: why small sample size undermines the reliability of neuroscience. *Nat. Rev. Neurosci.* 14(5):365, 2013.
- ²Debayle, J., Y. Gavet, and J. C. Pinoli. General adaptive neighborhood image restoration, enhancement and segmentation. In: International Conference Image Analysis and Recognition, Springer, Berlin, 2006, pp. 29–40.
- ³Engel, S. A., G. H. Glover, and B. A. Wandell. Retinotopic organization in human visual cortex and the spatial precision of functional MRI. *Cereb. Cortex (New York, NY: 1991)* 7(2):181–192, 1997.
- ⁴Fang, F., and S. He. Cortical responses to invisible objects in the human dorsal and ventral pathways. *Nat. Neurosci.* 8(10):1380, 2005.
- ⁵Goodfellow, I., J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, *et al.* Generative adversarial nets. In: Advances in Neural Information Processing Systems, 2014.
- ⁶Güçlütürk, Y., U. Güçlü, K. Seeliger, S. Bosch, R. van Lier, and M. A. van Gerven. Reconstructing perceived faces from brain activations with deep adversarial neural decoding. In: Advances in Neural Information Processing Systems, 2017, pp. 4246–4257.
- ⁷Horikawa, T., and Y. Kamitani. Generic decoding of seen and imagined objects using hierarchical visual features. *Nat. Commun.* 8:15037, 2017.
- ⁸Huang, W., H. Yan, R. Liu, L. Zhu, H. Zhang, and H. Chen. F-score feature selection based Bayesian reconstruction of visual image from human brain activity. *Neurocomputing* 316:202–209, 2018.
- ⁹Kiehl, K. A., K. R. Laurens, T. L. Duty, B. B. Forster, and P. F. Liddle. An event-related fMRI study of visual and auditory oddball tasks. *J. Psychophysiol.* 15(4):221, 2001.
- ¹⁰Ledig, C., L. Theis, F. Huszár, J. Caballero, A. Cunningham, *et al.* Photo-realistic single image super-resolution using a generative adversarial network. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 4681–4690.
- ¹¹Ma, Y., X. Gu, and Y. Wang. Histogram similarity measure using variable bin size distance. *Comput. Vis. Image Underst.* 114(8):981–989, 2010.
- ¹²Nguyen, H. V., and L. Bai. Cosine similarity metric learning for face verification. In: Asian Conference on Computer Vision, Springer, Berlin, 2010, pp. 709–720.
- ¹³Norman, K. A., S. M. Polyn, G. J. Detre, and J. V. Haxby. Beyond mind-reading: multi-voxel pattern analysis of fMRI data. *Trends Cogn. Sci.* 10(9):424–430, 2006.
- ¹⁴Pluim, J. P., J. A. Maintz, and M. A. Viergever. Mutual-information-based registration of medical images: a survey. *IEEE Trans. Med. Imaging* 22(8):986–1004, 2003.
- ¹⁵Reed, S., Z. Akata, X. Yan, L. Logeswaran, B. Schiele, and H. Lee. Generative adversarial text to image synthesis. 2016. arXiv preprint [arXiv:1605.05396](https://arxiv.org/abs/1605.05396).
- ¹⁶Seeliger, K., U. Güçlü, L. Ambrogioni, Y. Güçlütürk, and M. A. van Gerven. Generative adversarial networks for reconstructing natural images from brain activity. *NeuroImage* 181:775–785, 2018.
- ¹⁷Shen, G., T. Horikawa, K. Majima, and Y. Kamitani. Deep image reconstruction from human brain activity. *PLoS Comput. Biol.* 15(1):e1006633, 2019.
- ¹⁸St-Yves, G., and T. Naselaris. Generative adversarial networks conditioned on brain activity reconstruct seen images. In: 2018 IEEE International Conference on Systems, Man, and Cybernetics (SMC), IEEE, 2018, pp. 1054–1061.
- ¹⁹Wang, Z., A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli. Image quality assessment: from error visibility to structural similarity. *IEEE Trans. Image Process.* 13(4):600–612, 2004.
- ²⁰Yi, Z., H. Zhang, P. Tan, and Gong, M. Dualgan. Unsupervised dual learning for image-to-image translation. In: Proceedings of the IEEE International Conference on Computer Vision, 2017, pp. 2849–2857.
- ²¹Zhang, H., T. Xu, H. Li, S. Zhang, X. Wang, X. Huang, D. Metaxas, and N. Stackgan. Text to photo-realistic image synthesis with stacked generative adversarial networks. In: Proceedings of the IEEE International Conference on Computer Vision, 2017, pp. 5907–5915.

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